

#HW 8

● Graded

Student

Jewon Park

Total Points

100 / 100 pts

Question 1

(no title)

30 / 30 pts

✓ - 0 pts Correct

Question 2

(no title)

10 / 10 pts

✓ - 0 pts Correct

- 4 pts $\Delta M = \Delta t (M_{in} - M_{out})$

- 2 pts the calculation was asked for 1 hour, not 30 mins.

Question 3

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 3 pts $m_1 = m_2 + m_3$

- 3 pts $m_1 = 10.27 \text{ kg/s}$, $m_2 = m_3 = 5.13 \text{ kg/s}$

- 5 pts $1 \text{ bar} = 10^5 \text{ Pa}$; $m_1 = m_2 + m_3$ where, $m_2 = m_3$.

Question 4

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 3 pts $Q = m [C_p \Delta T + ((V_2^2 - V_1^2)/2)]$

- 5 pts rate of heat transfer not calculated/ not included in scan

- 4 pts $v_1 = 6.25 \text{ m/s}$; $v_2 = 7.37 \text{ m/s}$; $Q = 10.56 \text{ kW}$

- 2 pts $v_2 = 7.37 \text{ m/s}$

Question 5

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 5 pts $v_2 = \sim 149 \text{ ft/s}$; check the unit conversions carefully.

- 7 pts incomplete execution of solution

Question 6

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 3 pts 133.1 kW

- 5 pts 133.1 kW; check the calculation clearly.

Question assigned to the following page: [1](#)

CHE 201: Introduction to Thermodynamics

Expires Dec 29th, 2025



[< Back](#)

HW 8 Question 1

79 / 79 points

Due: 10/27/2025, 11:59 PM CDT

- | | | | |
|---|---|---|----------------|
|  |  | <u>3.13 Internal energy, enthalpy, and specific heats of ideal gases</u> | 8 / 8 points |
|  |  | <u>3.14 Applying the energy balance using ideal gas tables, constant specific heats, and software</u> | 10 / 10 points |
|  |  | <u>3.15 Polytropic process relations</u> | 10 / 10 points |
|  | | <u>7.1 Conservation of mass for a control volume</u> | 2 / 2 points |
|  | | <u>7.2 Forms of the mass rate balance</u> | 5 / 5 points |
|  | | <u>7.3 Applications of the mass rate balance</u> | 5 / 5 points |
|  | | <u>7.4 Conservation of energy for a control volume</u> | 2 / 2 points |
|  |  | <u>7.5 Nozzles and diffusers</u> | 10 / 10 points |
|  |  | <u>7.7 Turbines</u> | 10 / 10 points |
|  |  | <u>7.8 Analyzing control volumes at steady state</u> | 9 / 9 points |



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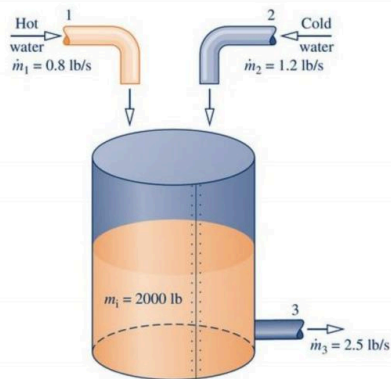


Assignments

Questions assigned to the following page: [2](#) and [3](#)

EXERCISE 7.18.6: (Problem 4.6 in the 9th edition).

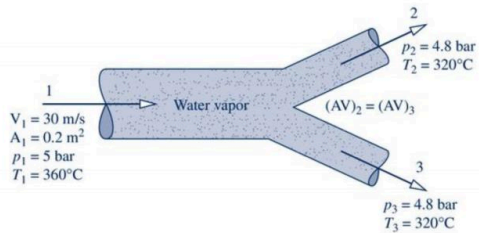
- (a) The figure below shows a mixing tank initially containing 2000 lb of liquid water. The tank is fitted with two inlet pipes, one delivering hot water at a mass flow rate of 0.8 lb/s and the other delivering cold water at a mass flow rate of 1.2 lb/s. Water exits through a single exit pipe at a mass flow rate of 2.5 lb/s. Determine the amount of water, in lb, in the tank after one hour.



$$\begin{aligned}\dot{m}_{in} - \dot{m}_{out} &= 0.8 + 1.2 - 2.5 = -0.5 \text{ lb/s} \\ m_0 &= 2000 \text{ lb} \\ m(t) &= 2000 - 0.5t \\ t &= 1 \text{ hr} = 3600 \text{ s} \\ m(3600) &= 2000 - 0.5(3600) = 200 \text{ lb} \\ t &= \frac{2000}{0.5} = 4000 \text{ s}, 4000 \text{ s} \Rightarrow 66.7 \text{ min} > 60 \text{ min} \\ \therefore &200 \text{ lb}\end{aligned}$$

EXERCISE 7.18.14: (Problem 4.14 in the 9th edition).

- (a) The figure below provides steady-state data for water vapor flowing through a piping configuration. At each exit, the volumetric flow rate, pressure, and temperature are equal. Determine the mass flow rate at the inlet and exits, each in kg/s.



$$\begin{aligned}P_1 &= 5 \text{ bar} & T_1 &= 360^\circ\text{C} \\ V_1 &= 0.57946 \text{ m}^3/\text{kg} \\ \dot{m} &= \frac{\dot{V}}{v} = \frac{AV}{v} \\ \dot{m}_1 &= \frac{0.2 \times 30}{0.57946} = 10.35 \text{ kg/s} \\ \dot{m}_2 &= \dot{m}_3 \\ \dot{m}_1 &= \dot{m}_2 + \dot{m}_3 = 2\dot{m}_{out} \\ \Rightarrow \dot{m}_2 = \dot{m}_3 &= \frac{\dot{m}_{in}}{2} = \frac{10.35}{2} = 5.18 \text{ kg/s}\end{aligned}$$

Questions assigned to the following page: [4](#), [5](#), and [6](#)



EXERCISE

7.18.18: (Problem 4.18 in the 9th edition).



- (a) Air enters a horizontal, constant-diameter heating duct operating at steady state at 290 K, 1 bar, with a volumetric flow rate of $0.25 \text{ m}^3/\text{s}$, and exits at 325 K, 0.95 bar. The flow area is 0.04 m^2 . Assuming the ideal gas model with $k = 1.4$ for the air, determine (a) the mass flow rate, in kg/s, (b) the velocity at the inlet and exit, each in m/s, and (c) the rate of heat transfer, in kW.

$$a) \rho = \frac{P}{RT} = \frac{100}{0.287 \times 290} = 1.202 \text{ kg/m}^3$$

$$\dot{m} = \rho \cdot \dot{V}_1 = 1.202 \times 0.25 = 0.3 \text{ kg/s}$$

$$b) V_1 = \frac{\dot{V}_1}{A} = \frac{0.25}{0.04} = 6.25 \text{ m/s}$$

$$\rho_2 = \frac{P_2}{RT_2} = \frac{0.95}{0.287 \times 325} = 1.02 \text{ kg/m}^3$$

$$V_2 = \frac{\dot{V}_2}{A} = \frac{0.295}{0.04} = 7.38 \text{ m/s}$$

$$c) \dot{Q} = \dot{m} \left[c_p (T_2 - T_1) + \frac{V_2^2 - V_1^2}{2} \right]$$

$$c_p (T_2 - T_1) = 1.0045 (325 - 290) = 35.16 \text{ kJ/kg}$$

$$\Delta KE = \frac{7.38^2 - 6.25^2}{2000} = 0.0077 \text{ kJ/kg}$$

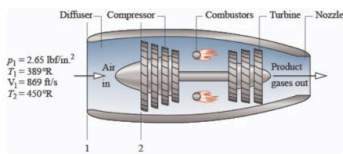
$$\dot{Q} = 0.3 \times (35.16 + 0.0077) = 10.6 \text{ kW}$$



EXERCISE

7.18.25: (Problem 4.25 in the 9th edition).

- (a) As shown in the figure below, air enters the diffuser of a jet engine operating at steady state at 2.65 lbf/in^2 , 389°R and a velocity of 869 ft/s , all data corresponding to high-altitude flight. The air flows adiabatically through the diffuser and achieves a temperature of 450°R at the diffuser exit. Using the ideal gas model for air, determine the velocity of the air at the diffuser exit, in ft/s.



$$h_1 + \frac{V_1^2}{2g_c} = h_2 + \frac{V_2^2}{2g_c} \Rightarrow c_p (T_1 - T_2) = \frac{V_2^2 - V_1^2}{2g_c}$$

$$V_2^2 = V_1^2 + 2g_c c_p (T_1 - T_2) = 869^2 + 2(32.174)(186.8)(389 - 450)$$

$$= 2.2 \times 10^4$$

$$V_2 = \sqrt{2.2 \times 10^4} = 149 \text{ ft/s}$$



EXERCISE

7.18.31: (Problem 4.31 in the 9th edition).



- (a) Steam enters a turbine operating at steady state with a mass flow of 10 kg/min , a specific enthalpy of 3100 kJ/kg , and a velocity of 30 m/s . At the exit, the specific enthalpy is 2300 kJ/kg and the velocity is 45 m/s . The elevation of the inlet is 3 m higher than at the exit. Heat transfer from the turbine to its surroundings occurs at a rate of $1.1 \text{ kJ per kg of steam flowing}$. Let $g = 9.81 \text{ m/s}^2$. Determine the power developed by the turbine, in kW.

$$\dot{W} = \dot{m} \left[(h_1 - h_2) + \frac{V_1^2 - V_2^2}{2 \times 1000} + g \frac{(z_1 - z_2)}{1000} - q_{out} \right]$$

$$h_1 - h_2 = 3100 - 2300 = 800 \text{ kJ/kg}$$

$$\frac{V_1^2 - V_2^2}{2000} = \frac{30^2 - 45^2}{2000} = -0.5625 \text{ kJ/kg}$$

$$g \frac{(z_1 - z_2)}{1000} = \frac{9.81 \times 3}{1000} = 0.0294 \text{ kJ/kg}$$

$$\Rightarrow \dot{W} = 0.6667 \times (800 - 0.5625 + 0.0294 - 1.1)$$

$$= 133.1 \text{ kW}$$